

Classification of Birds by Sound

By Jennifer Poling

Introduction

This project uses deep learning to classify Seattle birds by their sounds. The species of birds included in this project are the American Crow, American Robin, Bewick's Wren, Black-capped Chickadee, Dark-eyed Junco, House Finch, House Sparrow, Northern Flicker, Red-winged Blackbird, Song Sparrow, Spotted Towhee, and White-crowned Sparrow. The question being investigated is, "Can a convolutional neural network model predict the bird species from spectrograms of the bird's sounds?" This model can be used in ecology and environmental science.

A convolutional neural network (CNN) is built to predict which bird species made different sounds. The data used in this project is made up of two second long spectrograms of bird sounds labeled with the species of bird. The data comes from the birdcall competition data on Kaggle [1]. The creator of the page on Kaggle sourced the data from Xeno-Canto [2]. The data that I used was preprocessed and can be accessed from GitHub [3]. The test data can also be accessed from GitHub [4]. The code is located at <https://github.com/jnplng/bird-classification>.

Theoretical Background

The model used in this project is a convolutional neural network (CNN). A CNN is a deep learning model. Neural networks have an input layer. The number of features in the input layer is determined by the number of predictors in the data. There are weights that connect the input layer to the hidden layer. A bias term is added to the weighted sum of the neuron's inputs. The parameters of the weights and biases can be tuned as part of training the model.

A linear transformation is applied to the input by the weights. For each neuron, output is computed as a weighted sum of the inputs plus a bias term. This can be expressed using this equation: $z = b + \sum w_i x_i$. In this equation, w_i are the weights, x_i are the input features, and b is the bias. A nonlinear activation function is then applied to this result to produce the neuron's output, represented by this equation: $z = f(z)$. In this equation, f is an activation function such as sigmoid. Neural networks are made up of many layers, which each perform a linear transformation followed by a nonlinear transformation. Neural networks can model complex, nonlinear relationships between inputs and outputs because they stack these hidden layers.

There can be many parameters in a neural network; there are often more parameters than there is data. Therefore, neural networks are prone to overfitting. One way to combat overfitting is to use regularization. Dropout regularization is often used when building neural

networks. Dropout regularization tells the model to ignore a certain percentage of nodes in a layer.

Rectified linear unit, or ReLU, activation is a popular choice for the activation function in neural networks and was used in this project. Sigmoid activation was used for the binary model in this project. It is often used for binary projects because the results will be between zero and one, as is necessary for a binary model that is predicting either zero or one. Softmax activation was used for the multi-class model in this project because it causes all probabilities to add to one.

Convolutional neural networks are used for image classification. A CNN was used in this project because the audio data was turned into a spectrogram, which is a visual representation of the audio data. Spectrograms were used in this project because they provide a visual representation of auditory data while preserving the frequency and time-based information of the original audio data. A spectrogram is created by applying a Fourier transform to short windows of the audio signal. Each spectrogram for this project was generated from preprocessed audio clips that were subsampled to 22,050 Hz and segmented into short time windows, producing a 128 x 517 matrix. The x-axis represents time and the y-axis represents frequency. A spectrogram representation makes it possible to capture patterns in sound in a format similar to an image. Thus, a CNN could be used to classify the spectrogram image.

Convolutional neural networks use a convolutional filter in the convolution layer. Then each convolution layer is followed by a pooling layer. A convolutional filter basically slides a filter across an image, matching similar features in the image to features in the filter, and computes a smaller set of features using dot product multiplication. Hyperparameters that can be tuned in the convolution layer include the size of the filter and the number of filters applied. The size of the filter should be determined based on the size of the image so that the filter is much smaller than the image. The convolution layer produces a lot of data, and the pooling layer shrinks the data back to a smaller size.

The models in this project were evaluated using accuracy. This is because the models were classification models. The purpose of the models was to accurately identify the bird species from the spectrogram of the bird sounds.

Methodology

Each spectrogram for this project was generated from preprocessed audio clips that were subsampled to 22,050 Hz and segmented into short time windows, producing a 128 x 517 matrix. The x-axis represents time and the y-axis represents frequency. A spectrogram representation makes it possible to capture patterns in sound in a format similar to an image. Thus, a CNN could be used to classify the spectrogram image.

The data was normalized and split into train and test sets. Because the minimum value was -80, the data was normalized by dividing each item by 80. This meant that the normalized values were between negative one and approximately zero.

In order to create a binary model, the data for the Bewick's Wren and Spotted Towhee were separated from the rest of the data. These species were chosen because there was a similar amount of data for each of them. They were chosen to avoid a problem with class imbalance. After separating out the data for the Bewick's Wren and Spotted Towhee, the data set had 281 items.

Because the data set was small a small CNN model (the first binary model) was specified using three convolution layers with filters of size 16, 32, and 64 and a dropout rate of 0.5. The model was compiled using binary crossentropy for loss, since it's a binary model, optimizer type adam, and the metric accuracy. Accuracy was chosen as the metric because the classification model is predicting one species of bird or another, and the goal is accurate predictions. The model was trained with 15 epochs. Next the same model was used with more epochs and early stopping (the second binary model). Next the same model was used with higher patience in the early stopping settings (the third binary model). Next a larger CNN model was built. This model still had three convolution layers but with filters of size 32, 64, and 128. Next a smaller dropout rate was used on the smaller binary model.

The next portion of the project was to make a multi-class CNN. Labels were one-hot encoded. A CNN model was specified using three convolution layers with filters of size 32, 64, and 128, a dropout rate of 0.5, and 15 epochs (the first multi-class model). Next a smaller model was specified using three convolution layers with filters of size 32, 64, and 64, a dropout rate of 0.6, and 10 epochs (the second multi-class model). Next the smaller model was tried again but with a dropout rate of 0.5 (the third multi-class model). Then class weights were computed and used in training (the fourth multi-class model). Next the class weights were shifted halfway toward one rather than raw class weights (the fifth multi-class model).

F1 and Log Loss were computed on the top performing binary model and the top performing multi-class model.

Finally, the top performing model used to predict the bird species in the external test data. The external test data was split into two second long chunks of sound. These two second chunks were turned into spectrograms, which were normalized the same way the original data was normalized. Then the model predicted which species was represented by each spectrogram.

Results

The first binary model, using three convolution layers with filters of size 16, 32, and 64 and a dropout rate of 0.5, had a test accuracy of 73.68%. A confusion matrix of this model is included below in Figure 1.

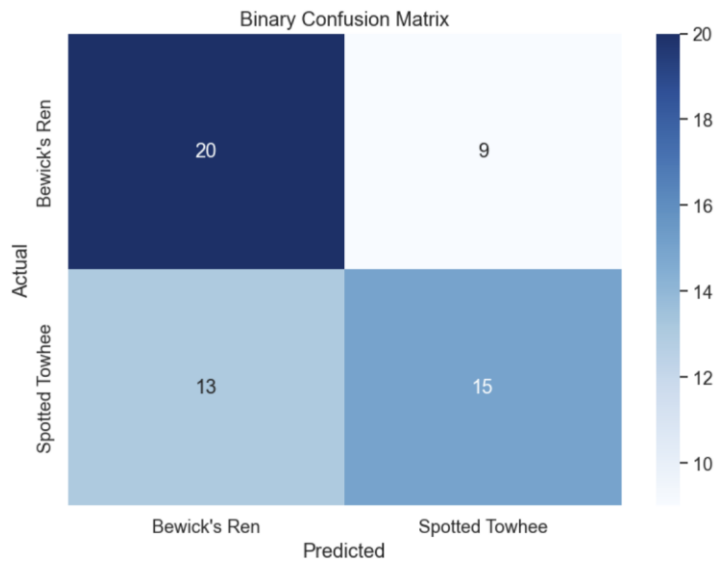


Figure 1 Confusion Matrix for Best Binary Model.

A plot of the binary model training versus validation accuracy over epochs is shown below in Figure 2. Training accuracy increases from approximately 0.48 to above 0.65. Validation accuracy drops sharply after the first epoch and remains lower, with only slight improvement in later epochs.

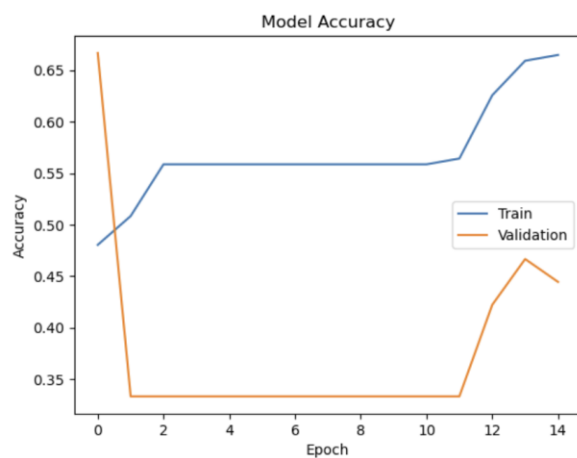


Figure 2 Binary Model Training Versus Validation Accuracy

The plot in Figure 3 below shows training and validation loss (log loss) over epochs. Training loss decreases steadily, indicating improved fit to the training data. In contrast, validation loss fluctuates and trends slightly increasing, remaining higher than the training loss for most of the training.

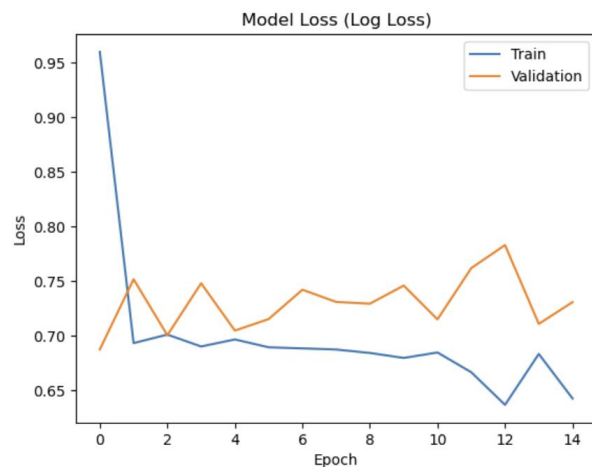


Figure 3 Binary Model Training Versus Validation Loss

The second binary model, which was the same model but with more epochs and early stopping, had a test accuracy of 49.12%.

The third binary model, which was the same as the second model but with higher patience, had a test accuracy of 61.40%.

The larger binary model had a test accuracy of 52.63%.

The small binary model with a smaller dropout rate had a test accuracy of 54.39%.

All of the model results are included in Figure 4 below.

Binary Model Results	
Model	Test Accuracy
small model	73.68%
early stopping	49.12%
higher patience	61.40%
smaller dropout	54.39%
large model	52.63%

Figure 4 Binary Model Accuracy Results

The first multi-class model had a test accuracy of 42.82%. The second multi-class model had a test accuracy of 43.58%. The third multi-class model had a test accuracy of 45.59%. The model using raw class weights had a test accuracy of 7.30% (the fourth multi-class model). The fifth

multi-class model using adjusted class weights had a test accuracy of 44.08%. All of the multi-class model results are included in Figure 5 below.

Multi-Class Model Results	
Model	Test Accuracy
base model	42.82%
smaller model	43.58%
smaller dropout	45.59%
raw class weights	7.30%
adjusted class weights	44.08%

Figure 5 Multi-Class Model Accuracy Results

The confusion matrix for the top performing multi-class model is included in Figure 6 below.

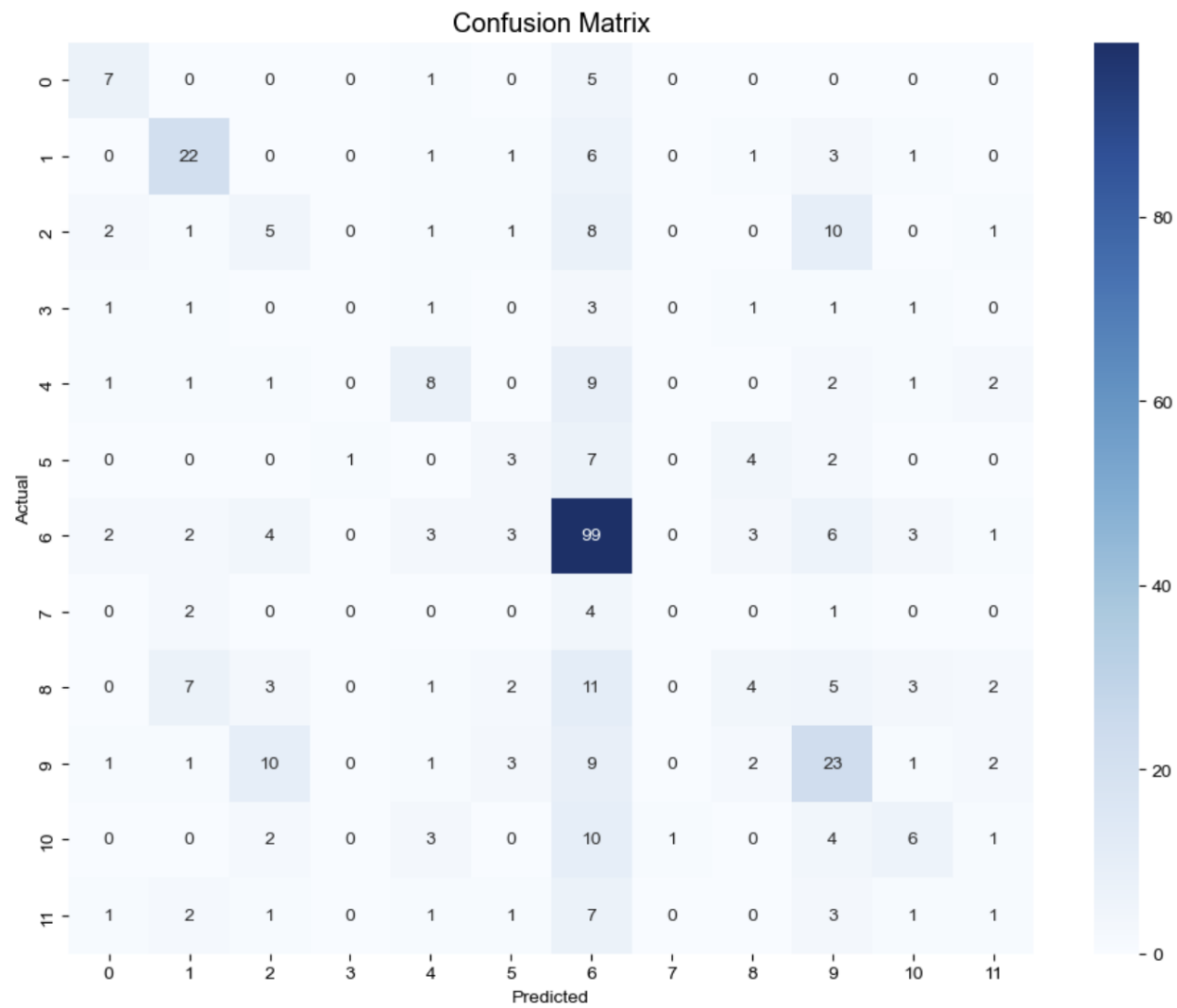


Figure 6 Multi-Class Confusion Matrix

Confusion Matrix Legend	
Bird Species	Number
American Crow	0
American Robin	1
Bewick's Ren	2
Black-capped Chickadee	3
Dark-eyed Junco	4
House Finch	5
House Sparrow	6
Northern Flicker	7
Red-winged Blackbird	8
Song Sparrow	9
Spotted Towhee	10
White-crowned Sparrow	11

Figure 7 Multi-Class Model Confusion Matrix Legend

A plot of the multi-class model training versus validation accuracy over epochs is shown below in Figure 8. Training accuracy increases steadily. Validation accuracy also improves over epochs, although it is lower than training accuracy as the number of epochs increases.

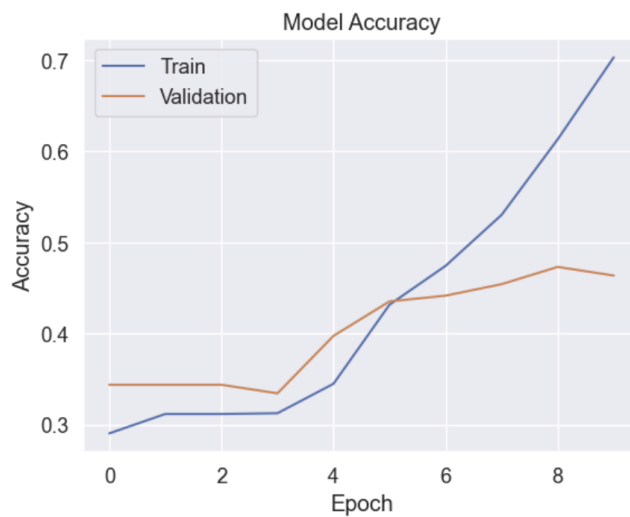


Figure 8 Multi-Class Model Training Versus Validation Accuracy

Figure 9 below shows training and validation log loss for the best multi-class model. Training loss decreases over the epochs, indicating improved fit to the training data. Validation loss decreases initially then stabilizes and increases slightly in later epochs.



Figure 9 Multi-Class Model Training Versus Validation Loss

F1 and Log Loss were computed on the top performing binary model and the top performing multi-class model. For the binary model, F1 was approximately 0.5769 and Log Loss was approximately 0.6741. For the multi-class model, Macro F1 was approximately 0.2618, and Log Loss was approximately 2.0527.

The top performing multi-class model was used to predict the bird species represented in the external test data. The model predicted that the first external test data included sounds of the House Sparrow and the Red-Winged Blackbird.

Chunk 0: House Sparrow (0.51)

Chunk 1: House Sparrow (0.48)

Chunk 2: House Sparrow (0.35)

Chunk 3: House Sparrow (0.45)

Chunk 4: Red-Winged Blackbird (0.48)

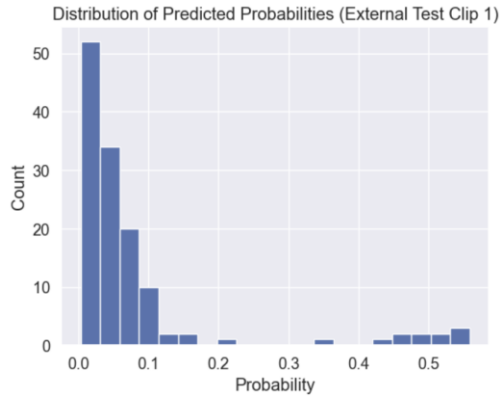


Figure 10 Distribution of Predicted Probabilities for Test Clip 1

The model predicted that the second external test data included sounds of only the Red-Winged Blackbird.

Chunk 0: House Sparrow (0.67)

Chunk 1: House Sparrow (0.63)

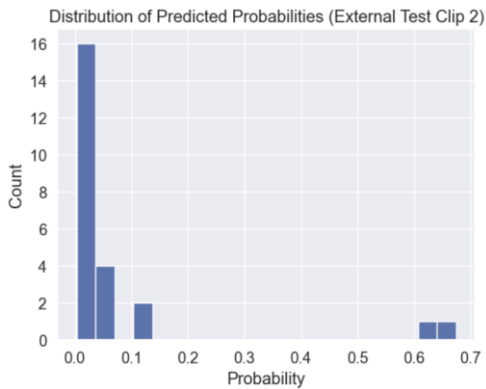


Figure 11 Distribution of Predicted Probabilities Test Clip 2

The model predicted that the third external test data included sounds of the House Sparrow and the Black-capped Chickadee.

Chunk 0: House Sparrow (0.24)

Chunk 1: House Sparrow (0.57)

Chunk 2: House Sparrow (0.26)

Chunk 3: House Sparrow (0.57)

Chunk 4: Black-capped Chickadee (0.36)

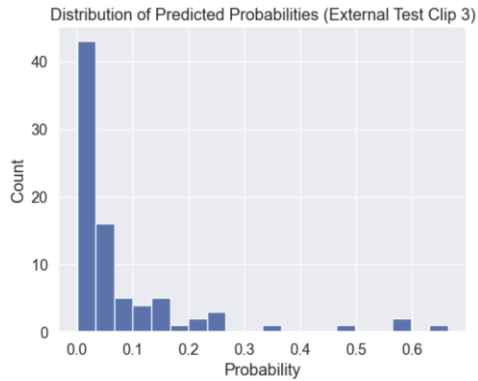


Figure 12 Distribution of Predicted Probabilities Test Clip 3

Discussion

The small binary model has a strong test accuracy of 73.68%. It looked like the training accuracy might be able to improve with more epochs. At the same time, the loss was decreasing. The training versus validation accuracy plot suggests overfitting. The divergence between training and validation loss also suggests overfitting. Therefore, the same model was tried again with and early stopping. That model finished after four epochs, possibly because the data set was small and the validation metrics were noisy. This was why a higher patience setting was used in the early stopping metrics. The test accuracy improved, but it was not as high as the test accuracy for the original model.

The larger binary model overfit the small data set, which resulted in a lower test accuracy. The smaller binary model was determined to be the stronger model. Next, a smaller dropout rate was used with the smaller model. The smaller dropout rate did not improve the model, likely due to the small data set.

The original small model was determined to be the strongest binary model. The F1 score indicates moderate performance in distinguishing between the two bird species. This suggests that it captures some meaningful patterns in the spectrogram data but still makes a significant number of classification errors. The log loss indicates that the model lacks strong confidence in its predictions.

The first multi-class model appeared to be overfitting. This could be because multi-class classification is harder for the model to learn. The second model with a higher dropout and fewer epochs appeared to be underfitting, so the next model was kept small but again used the original dropout rate. The next model had class weights computed and used in training to try to account for the class imbalance. Raw class weights caused what seems to be random guessing by the model, so next changing the formula for the class weights was tried. The top performing

model was the smaller model with the 0.5 dropout rate. However, the low macro F1 and high log loss indicate that even the best model does not generalize well to unseen data. It struggles to distinguish between bird species and does not produce strong predictions. The training and validation curves for this model suggest moderate overfitting. The gap between them indicates that the model fits the training data more effectively than unseen data. The loss curves indicate overfitting. While the model continues to reduce error on the training data, validation loss stops improving. This suggests that the model is learning patterns specific to the training data rather than generalizing. This is consistent with the low macro F1 score and high log loss.

The distribution of predicted probabilities for the external test clips show that most probabilities are near zero, with relatively few values approaching higher confidence levels. The model generally assigns low probability to most classes and has low confidence in its predictions. The absence of high probability values suggests that the model struggles to clearly distinguish between bird species.

One of the limitations in this homework was a lack of prior experience using HDF5 format. It took time to learn how to access that data. The models didn't take an unreasonable amount of time to train. The longest training time experienced was just over three minutes. This could be considered a long time, but the assignment gave the impression that it would take time to train the models. It also wasn't a long time compared to the wait time during big data systems assignments.

Another limitation was the model results changing. When the code was rerun after the report had been written, the test accuracy results changed. The Jupyter notebook code¹ is included in the GitHub because this notebook has the test accuracy results used in the report. The first binary model was rerun to make a confusion matrix. Because the same name was used for all of the models, the model had to be specified again and refit. The test accuracy was much lower in this later running. After running again, the new test accuracy was closer to the original, but still not as good. The process can be followed in the Jupyter notebook code. The other Jupyter notebook is just provided for the original test accuracy results. For future models, setting an overall seed at the beginning might help prevent this. However, part of the issue is due to the nature of convolutional neural networks and the fact that a small data set was used. The network may learn slightly different things when it runs than it had the previous time, especially when the data set is small.

The model often predicted Bewick's Wren when the bird was actually the Song Sparrow. Examination of the spectrograms showed similarities between the two species, particularly in the frequency range and the presence of repeating vertical chirp patterns. Many of the misclassified Song Sparrow spectrograms appeared blurry or contained dense patterns that were difficult to distinguish clearly. Also, some Song Sparrow spectrograms exhibited vertical

banding and overlapping features that resembled patterns found in Bewick's Wren examples. These similarities in time-frequency structure likely contributed to the model confusing the two species.

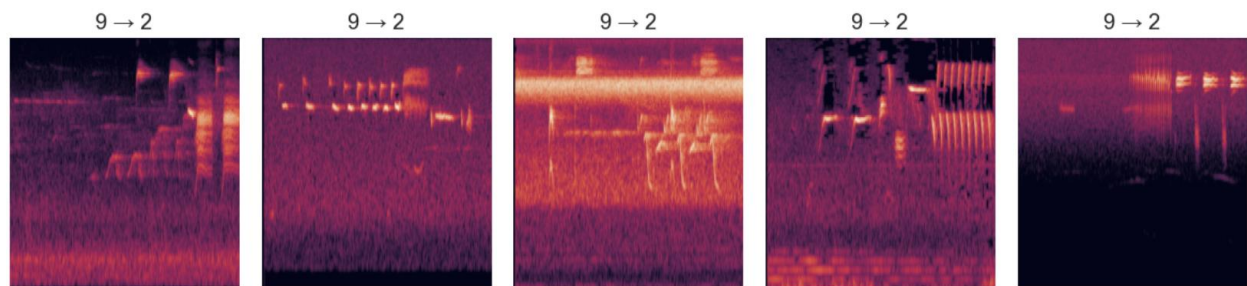


Figure 13 Bewick's Wren Spectrograms Misclassified as Song Sparrow

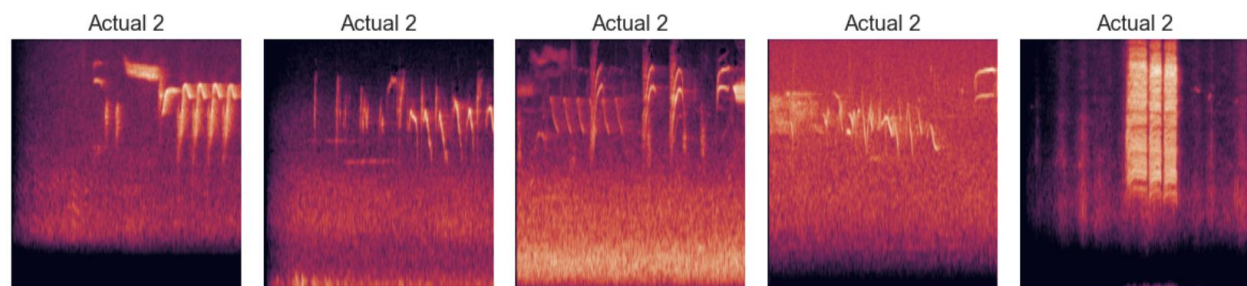


Figure 14 Song Sparrow Spectrogram Samples

Other models that could have been used to perform this task include random forest and support vector machines, but they would have required more feature engineering. A neural network makes sense for this task because this is a task that convolutional neural networks are very good at.

Better results could have been obtained using a pre-trained model. Companies who have made pre-trained models available put a lot of time, money, and expertise into creating these models. Students building from scratch are unlikely to match the results they could obtain building on a pre-trained model.

Conclusions

The purpose of this project was to determine whether a convolutional neural network could predict bird species using spectrograms of the birds' sounds. The models built during this project struggled to predict the bird species correctly. One way to build on this project would be to use a pre-trained model as the basis for a model to predict bird species. This approach is likely to produce a model that is more successful at predicting bird species. Another way to

build on this project would be to collect more data. More data, especially more balanced data, could produce a more robust model.

There are some broader impacts and real-world implications of this work. A strong model that is good at predicting bird species from their sounds could be used in environmental work. It could be used to track the types of birds in an area. Birds can be hard to spot in trees and bushes. It can be easier to make recordings of the birds in an area to determine which birds are there. A strong model could also be used to measure the health of a habitat by determining the number of bird species present.

References

- [1] Vopani, "Xeno-Canto Bird Recordings Extended (A-M)," Kaggle, n.d. [Online]. Available: <https://www.kaggle.com/datasets/rohanrao/xeno-canto-bird-recordings-extended-a-m> (accessed May 19, 2026).
- [2] Xeno-Canto Foundation, "Xeno-Canto: Sharing Wildlife Sounds from Around the World," Xeno-Canto. <https://xeno-canto.org/> (accessed May 24, 2026).
- [3] A. Mendible, "bird_spectrograms.hdf5," GitHub, n.d. [Online]. Available: https://github.com/mendible/5322/blob/main/Homework%203/bird_spectrograms.hdf5 (accessed May 19, 2026).
- [4] A. Mendible, "test_birds," GitHub, n.d. [Online]. Available: https://github.com/mendible/5322/tree/51bd4705bf06d4f46e1848f9261da9b2db3333c0/Homework%203/test_birds (accessed May 19, 2026).